

Module 24.10: Lasso Regression Implementation & Shrinkage Paths

1. End-to-End Implementation Workflow

Applying regularized regression in practice requires three mandatory preprocessing prerequisites:

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| 1. Categorical Dummy Encoding: pd.get_dummies(df, drop_first=True)
| 2. Train-Test Split: 80% Training, 20% Unseen Testing Data
| 3. Mandatory Standardization: StandardScaler() fitted STRICTLY on training split
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Why Feature Scaling is Mandatory for Regularization:

Regularization applies a uniform penalty $\lambda|w_j|$ across all parameters.

- If Feature A is measured in dollars ($\$10^6$) and Feature B in age ($\10^1), the unscaled weight w_A is naturally tiny while w_B is large.
- The regularizer would penalize w_B severely simply because of its arbitrary physical unit!
- Standardization brings all features to mean 0 and variance 1, ensuring fair, scale-invariant shrinkage.

2. Tracing the Regularization Path

A **regularization path** plots parameter weights w_j on the vertical axis against the regularization strength λ (or $\log(\alpha)$) on the horizontal axis:

- As $\lambda \rightarrow 0$: Weights equal the unconstrained Ordinary Least Squares solution.
- As $\lambda \rightarrow \infty$: All weights are progressively forced to zero ($w_j = 0$).
- The order in which coefficients hit zero reveals their relative empirical importance.

